Developping simulation and statistical evaluation tools for a non-linear mixed-effect models

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Abstract

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Neurodegenerative diseases are characterized by a slow and complex evolution, involving numerous heterogeneous clinical and biological markers (biomarkers, imaging, clinical scores) that evolve jointly over time. Modeling and predicting this progression from multivariate longitudinal data is a central challenge for clinical research and precision medicine.

In this context, Leaspy (LEArning Spatiotemporal Patterns in Python), an open-source Python library, was developed by the ARAMIS team at the Paris Brain Institute (ICM) in collaboration with INRIA, INSERM, CNRS and Sorbonne University to analyze the joint evolution of numerous markers simultaneously. The repository is available at `https://github.com/aramis-lab/leaspy` and the documentation at `https://leaspy.readthedocs.io/en/stable/index.html`.

The package is designed for the statistical analysis of longitudinal medical data consisting of repeated observations of patients at different time points. Its principal use cases include: reconstructing the long-term spatiotemporal trajectory of disease evolution from short-term cohort data; positioning individual patients relative to a group-average disease timeline; quantifying the impact of cofactors (sex, genetic mutations, environmental factors) on progression; imputing missing values; predicting future observations; and simulating virtual patient cohorts.

2.1 Mathematical Background

2.1.1 Nonlinear Mixed-Effects Models

Leaspy is based on the non-linear mixed-effects models framework, which decomposes variability in longitudinal observations into two complementary components [9]:

- **Fixed effects** (population-level): systematic trends shared across all individuals, such as the average disease trajectory, typical progression speed, and population reference time.
- **Random effects** (individual-level): subject-specific deviations capturing heterogeneity in onset time, progression speed, and the relative ordering of biomarker changes.

Formally, let y denote the observed data, $z = (z_{\text{fe}}, z_{\text{re}})$ the latent fixed and random effects, θ the model parameters, and Π the known hyperparameters. Parameter estimation is performed by maximizing the marginal likelihood

$$p(y \mid \theta, \Pi) = \int_z p(y, z \mid \theta, \Pi)\, dz, \tag{1}$$

which decomposes in log-space as

$$\log p(y, z \mid \theta, \Pi) = \underbrace{\log p(y \mid z, \theta, \Pi)}_{\text{data attachment}} + \underbrace{\log p(z_{\text{re}} \mid z_{\text{fe}}, \theta, \Pi) + \log p(z_{\text{fe}} \mid \theta, \Pi)}_{\text{prior attachment}}. \tag{2}$$

Because this integral is intractable in closed form, Leaspy employs the MCMC-SAEM algorithm (Markov Chain Monte Carlo Stochastic Approximation Expectation-Maximization) [4]. This algorithm alternates between (i) a stochastic approximation step that uses Gibbs sampling to draw latent variables conditional on current parameter estimates, and (ii) a maximization step that re-estimates fixed effects and variance components from the updated sufficient statistics.

2.1.2 Riemannian Geometry of Disease Progression

A key innovation of Leaspy is the interpretation of disease progression within a Riemannian geometric framework [9]. The core idea is to represent the multivariate progression of clinical markers as a geodesic on a product Riemannian manifold, where the shape of the curve is entirely determined by the choice of the metric $G(p)$.

For clinical scores exhibiting floor or ceiling effects, common in neurodegenerative disease assessment, the manifold $(0, 1)$ equipped with the metric $G(p) = 1/[p^2(1 - p)^2]$ yields logistic (sigmoidal) trajectories. This metric encapsulates the bounded nature of the scores in a geometrically principled way. More generally, the Euclidean metric on $\mathbb{R}^n$ yields linear trajectories, allowing the framework to accommodate diverse data types.

The average population trajectory γ_0 is a geodesic parameterized by its initial conditions at reference time t_0: initial position $p_0 = \gamma_0(t_0)$ and initial speed $v_0 = \dot{\gamma}_0(t_0)$. Individual trajectories $\gamma_i(t)$ are derived from this average through three types of random effects:

- **Time shift** τ_i: earlier or later disease onset relative to the population average.
- **Log-acceleration factor** ξ_i: faster or slower progression speed.
- **Space shifts** $w_{i,k}$: modifications to the relative ordering and timing of individual biomarker progressions, encoded as tangent-space perturbations.

These three effects are combined through a latent disease age reparametrization:

$$\psi_i(t) = v_0\, e^{\xi_i}(t - \tau_i) + t_0, \tag{3}$$

which transforms the calendar age t of patient i into a disease-stage age. This reparametrization provides a biologically interpretable decomposition: τ_i captures onset heterogeneity, e^{ξ_i} captures progression rate heterogeneity, and the space shifts $w_{i,k} = (\mathbf{A}\mathbf{s}_i)_k$, obtained via the mixing matrix $\mathbf{A}$ acting on independent sources $\mathbf{s}_i$, capture differences in the sequential ordering of biomarker changes across patients.

2.2 Main Models

Logistic Model. The logistic model is the core model of Leaspy, designed for multivariate longitudinal data whose components exhibit sigmoidal (bounded, monotone) progressions, typical of cognitive scores or normalized biomarkers.

Joint Model. The joint model [7] extends the logistic model to simultaneously analyze longitudinal measurements and time-to-event data (survival, clinical events, competing risks). Rather than treating these processes separately, which can lead to biases from informative dropout, the joint model links them through the shared latent disease age $\psi_i(t)$, avoiding the need to define an explicit disease reference time.

Mixture Model. Standard mixed-effects models assume a single population trajectory with random individual deviations. The mixture model [1] relaxes this assumption by modeling the population as a combination of n_c latent subgroups, each with its own distribution of individual parameters. This formulation enables simultaneous disease progression modeling and probabilistic patient subtyping.

2.3 Main Algorithms

Leaspy provides four main algorithmic operations:

`fit`: Estimates population parameters (fixed effects and variance components) from a cohort of longitudinal observations using MCMC-SAEM. The user specifies the model type, number of outcomes, number of latent sources, and observational noise model. The result is a fitted model characterizing the average disease trajectory and its population-level variability.

`personalize`: Given a fitted model, estimates individual random effects $(\tau_i, \xi_i, \mathbf{s}_i)$ for each subject from their available observations. Two approaches are available: a frequentist mode using `scipy.optimize.minimize` to maximize the individual likelihood, and a Bayesian mode using a Gibbs sampler to extract the mean or mode of the posterior distribution. The output is a set of individual parameters that characterize each patient's position on the disease timeline.

`estimate`: Evaluates the modeled trajectory $\gamma_{i,k}(t)$ for a specific patient at arbitrary time points, using the previously estimated individual parameters. This enables reconstruction of the full disease trajectory, imputation of missing values, and forecasting of future clinical scores.

`simulate`: Generates synthetic patient cohorts by (i) sampling individual parameters from the learned population distributions, (ii) generating visit times according to user-specified schedules (random, regular, or custom), and (iii) evaluating model trajectories with added observational noise. This functionality is only available for the logistic model for now, but a part of this work consists in extending it to the joint model.

2.4 Applications and Related Work

Leaspy has been applied to neurodegenerative diseases, demonstrating its generalizability and clinical relevance.

Parkinson's disease. Leaspy comes with a built-in Parkinson's disease dataset, enabling out-of-the-box demonstration and benchmarking. Figure 1 illustrates the typical results obtained with the software.

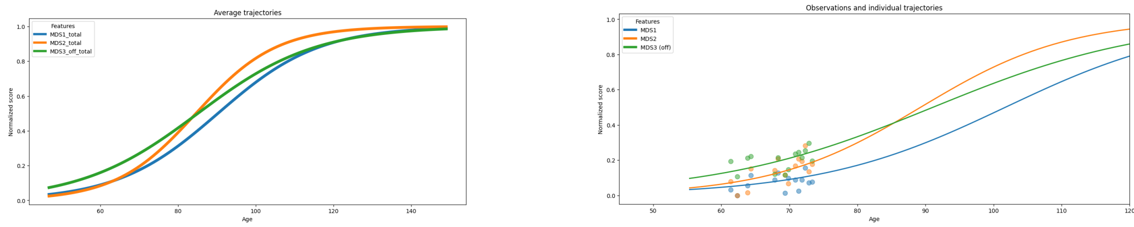

Figure 1: Disease progression modeling with Leaspy on the built-in Parkinson's dataset. **Left:** Average population trajectory of clinical scores over the estimated disease timeline. **Right:** Personalized trajectory (solid line) and observed visits (dots) for one patient.

Alzheimer's disease (AD). The AD Course Map [2] uses Leaspy to create a spatiotemporal atlas of AD progression by jointly modeling neuropsychological and imaging data. It outperformed 56 competing methods in the TADPOLE open challenge and, when tested globally on over 4,600 patients, demonstrated the ability to enrich clinical trials with predicted progressors, reducing required sample sizes by up to 50% [5].

Huntington's disease. Koval et al. applied the disease course mapping framework to identify the optimal intervention window for Huntington's disease clinical trials, demonstrating that the model could "spot the time to treat" by positioning patients on the disease timeline prior to symptom onset [3].

CADASIL. Kaisaridi et al. applied Leaspy to 395 patients from the French CADASIL referral center, modeling the joint progression of 14 clinical scores. Using a mixture model, they identified two clinically meaningful subgroups, an early-onset, rapidly progressing group and a late-onset, slowly progressing group, providing new insights into disease heterogeneity for this rare small vessel disease [1].

Amyotrophic lateral sclerosis (ALS). The joint model was validated on data from patients with ALS from the PRO-ACT dataset. The spatiotemporal model links longitudinal clinical decline to time-to-ventilation and survival events, outperforming standard shared-random-effect joint models in accuracy and discriminative ability while using fewer parameters [7].

3 Simulating Synthetic Patient Cohorts with the Joint Model

3.1 Details on the Joint Model

The joint model [7] extends the Leaspy longitudinal framework to simultaneously model repeated clinical measurements and survival outcomes, without requiring a reliable reference time point. This is a key advantage in neurodegenerative disease contexts where disease onset is not directly observable.

The model is composed of three interconnected components. The longitudinal sub-model describes the evolution of a clinical score via a logistic trajectory parameterized by the latent disease age $\psi_i(t)$ (Equation (3)), as already introduced in the main Leaspy framework. The survival sub-model describes the probability that patient i survives beyond time t using a Weibull distribution applied to the latent disease age:

$$S_i(t) = S_0(\psi_i(t)) = \mathbf{1}_{\psi_i(t)>t_0} \exp\left(- \left(\frac{\psi_i(t) - t_0}{\nu} \right)^{\rho} \right) + \mathbf{1}_{\psi_i(t)\leq t_0}, \tag{4}$$

where $\nu > 0$ is a scale parameter and $\rho > 0$ is the shape parameter of the Weibull distribution. The linkage structure connecting the two sub-models is the shared latent disease age $\psi_i(t) = e^{\xi_i}(t - \tau_i) + t_0$, which encodes individual variability in disease onset (τ_i) and progression rate (e^{ξ_i}). This differs from classical shared-random-effects joint models, which link the survival hazard to the value of the longitudinal outcome; here the link is structural, operating directly at the level of the disease timeline.

The joint log-likelihood decomposes additively into longitudinal and survival data attachment terms plus prior attachment terms for both fixed and random effects, under the assumption that survival and longitudinal processes are conditionally independent given the latent random effects [8].

Estimation proceeds via the MCMC-SAEM algorithm, initialized from a pre-trained longitudinal model and a Weibull Accelerated Failure Time (AFT) survival model. The algorithm runs for a burn-in phase followed by a Robbins-Monro convergence phase to extract posterior means of the fixed effects. Individual random effects (τ_i, ξ_i) are personalized in a second step by maximizing the posterior likelihood given each patient's observations and censored event time, using `scipy.optimize.minimize`.

The joint model was validated on simulated data following the ADEMP framework [6]. Using $M = 100$ datasets of $N = 200$ patients each, generated under the model's own structure with ALS-calibrated parameters, the estimation procedure recovered the true model parameters with low relative bias (under 5% for most parameters) and accurate coverage rates. Individual random effects were also recovered reliably, with high intraclass correlations between estimated and true values. Benchmarked against four reference models on real ALS data from the PRO-ACT dataset, an AFT survival model, the standalone longitudinal model, a two-stage model, and a shared-random-effects joint model (JMbayes2), the joint model achieved superior or comparable predictive and discriminative performance while using fewer parameters. Full details of the simulation study and benchmark results are reported in [7].

3.2 The Joint Simulation Algorithm

3.2.1 The Original Simulation Algorithm

In the original article describing the joint model [7], the simulation of synthetic patient cohorts under the joint model proceeds in six sequential steps, described below. This procedure generalizes the simulation workflow available in Leaspy for the logistic model to the joint setting.

1. **Simulate individual random effects**. For each patient, draw the log-acceleration factor ξ_i and time shift τ_i independently from their population distributions: $\xi_i \sim \mathcal{N}(0,\, \sigma_\xi^2), \quad \tau_i \sim \mathcal{N}(t_0,\, \sigma_\tau^2)$.

2. **Simulate age at first visit**. The age at first visit is drawn as $t_{i,0} = \tau_i + \delta_{f_i}$, where $\delta_{f_i} \sim \mathcal{N}(\bar{\delta}_f, \sigma_{\delta_f}^2)$ is a random offset relative to the patient's estimated disease onset.

3. **Simulate visit schedule**. Inter-visit intervals $\delta_{v_{i,j}} \sim \mathcal{N}(\bar{\delta}_v, \sigma_{\delta_v}^2)$ and a total follow-up duration $T_{f_i} \sim \mathcal{N}(\overline{T_f}, \sigma_{T_f}^2)$ determine the visit times $t_{i,0} < t_{i,1} < \cdots < t_{i,n_i}$, where n_i is the number of visits satisfying $t_{i,j} \leq t_{i,0} + T_{f_i}$.

4. **Simulate longitudinal outcomes**. The latent disease age $\psi_i(t_{i,j})$ is computed for each visit. The expected outcome value is then $\gamma_0(\psi_i(t_{i,j}))$ as defined by the logistic trajectory. Observed outcomes are sampled with additive noise; in the simulation study, a Beta distribution with mode $\gamma_0(\psi_i(t_{i,j}))$ and concentration p is used to match the bounded nature of clinical scores: $y_{i,j} \sim \mathcal{B}(\gamma_0(\psi_i(t_{i,j})), p)$.

5. **Simulate event times**. The event time T_{e_i} is drawn from the Weibull survival distribution implied by the latent disease age. Using the inverse-transform method, this is equivalent to: $T_{e_i} \sim e^{-\xi_i}\, \mathcal{W}(\nu, \rho) + \tau_i$, where $\mathcal{W}(\nu, \rho)$ denotes the Weibull distribution with scale ν and shape ρ.

6. **Apply censoring**. The event time and visit schedule are reconciled by applying two censoring rules: (i) all visits occurring after the event ($t_{i,j} > T_{e_i}$) are removed from the follow-up; (ii) events occurring after the last remaining visit ($T_{e_i} > t_{i,\max(j)}$) are treated as right-censored. This produces the observed tuple

$(y_{i,j}, T_{e_i}, B_i^e)$, where the censoring indicator $B_i^e = 1$ if the event was observed and $B_i^e = 0$ if it was censored.

The simulation algorithm relies on two disjoint sets of parameters. The model parameters, read directly from the fitted joint model, are: the population distributions of the individual random effects (t_0, σ_τ, σ_ξ), the Weibull survival parameters (ν, ρ), and the source-to-survival mixing weights ζ whenever the model includes latent sources. These parameters are set once and for all during model fitting and are not accessible to the user at simulation time. The study parameters, supplied by the user, govern the structure of the synthetic cohort: the number of patients N, the mean and standard deviation of the first-visit offset from disease onset ($\bar{\delta}_f$, σ_{δ_f}), the mean and standard deviation of the per-patient follow-up duration ($\overline{T_f}$, σ_{T_f}), and the mean and standard deviation of inter-visit intervals ($\bar{\delta}_v$, σ_{δ_v}). An additional technical parameter, `min_spacing_between_visits`, enforces a minimum gap between successive visits and prevents numerical artefacts from excessively close observations.

When the user does not supply some or all study parameters, they can instead provide a dataset, usually the same dataset used to fit the model. In that case, the missing parameters are estimated automatically from the observed visit process: $\bar{\delta}_f$ and σ_{δ_f} are derived from the distribution of first-visit ages (corrected by t_0 to express the offset from disease onset), $\overline{T_f}$ and σ_{T_f} are obtained from the empirical distribution of per-patient follow-up durations (last minus first visit), and $\bar{\delta}_v$ and σ_{δ_v} are estimated from the distribution of all observed inter-visit gaps. This feature is not present in the base simulation algorithm of Leaspy for the logistic model, but it is worth mentioning that it introduces a dependency on the training data and breaks the strict separation between the generative mechanism and the observed sample.

A critical artefact of the original algorithm. When the procedure described in [7] is applied without modification, a severe distributional mismatch appears in the simulated cohort. Specifically, a large fraction of simulated patients exhibit zero or near-zero follow-up and only a single observed visit. The root cause lies in the ordering of steps 3 and 5: the visit schedule is constructed first (steps 2–3), then the event time is sampled independently (step 5), and finally step 6 discards all visits occurring after the event. For patients with a short Weibull event time T_{e_i}, it is possible that $T_{e_i} < t_{i,0}$, meaning the event occurs before even the first planned visit. In that case, all visits are removed and the patient contributes no longitudinal observations to the simulated dataset. Because the Weibull distribution can generate short event times with non-negligible probability, especially for patients with large acceleration e^{ξ_i}, this mechanism inflates the mass near zero follow-up in a way that does not reflect the training cohort, where such patients are systematically under-represented due to study inclusion criteria.

3.2.2 The Anchor-Based Simulation Algorithm

To eliminate this artefact while preserving the marginal distributions of all simulated quantities, the implementation adopts an anchor-based construction of the visit schedule. The procedure differs from the original in steps 2–5 and proceeds as follows.

1. **Simulate individual random effects** (unchanged). $\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$ and $\tau_i \sim \mathcal{N}(t_0, \sigma_\tau^2)$.
2. **Simulate the event time first**. Before any visit is placed, draw the event time from its Weibull distribution: $T_{e_i} \sim e^{-\xi_i}\, \mathcal{W}(\nu, \rho) + \tau_i$.
3. **Simulate the study window independently of the event**. Draw a follow-up duration $T_{f_i} \sim |\mathcal{N}(\overline{T_f}, \sigma_{T_f}^2)|$, with an absolute value to ensure positivity, and a first-visit offset $\delta_{f_i} \sim \mathcal{N}(\bar{\delta}_f, \sigma_{\delta_f}^2)$. The natural study-end for patient i is $S_i = \tau_i + \delta_{f_i} + T_{f_i}$.
4. **Define the anchor**. Set the end of the observation window as
$$A_i = \begin{cases} T_{e_i} & \text{if } T_{e_i} \leq S_i \quad \text{(event observed)}, \\ S_i & \text{if } T_{e_i} > S_i \quad \text{(event right-censored)}. \end{cases}$$
5. **Build the visit schedule forward from $A_i - T_{f_i}$**. The baseline visit is placed at $t_{i,0} = A_i - T_{f_i}$, and successive visits are added by drawing inter-visit gaps $\delta_{v_{i,j}} \sim \mathcal{N}(\bar{\delta}_v, \sigma_{\delta_v}^2)$ until the next visit would exceed A_i. Only visits satisfying $t_{i,j} \leq A_i$ are retained.
6. **Simulate longitudinal outcomes and record the event** (unchanged). Outcomes $y_{i,j} \sim \mathcal{B}(\gamma_0(\psi_i(t_{i,j})), p)$ and the event tuple (T_{e_i}, B_i^e) are recorded as before.

With this new procedure, the visit window $[A_i - T_{f_i},\, A_i]$ always has length exactly T_{f_i}, regardless of whether the event is early or late. Every patient therefore contributes at least one baseline visit and a follow-up spanning T_{f_i} time units, which matches the empirical distribution of the training cohort. The artefactual mass at zero follow-up is eliminated without altering the marginal distributions of T_{e_i}, τ_i, or ξ_i.

3.3 Results and Validation

3.3.1 Using Simulated Data

3.3.2 Using Models Fitted on Real Data

4 TODO

5 Conclusion

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References

[1] Sofia Kaisaridi, Juliette Ortholand, Caglayan Tuna, Nicolas Gensollen, and Sophie Tezenas Du Montcel. A mixture model for subtype identification: Application to cadasil. In *ISCB 46-46th Annual Conference of the International Society for Clinical Biostatistics*, 2025.

[2] Igor Koval, Alexandre Bône, Maxime Louis, Thomas Lartigue, Simona Bottani, Arnaud Marcoux, Jorge Samper-Gonzalez, Ninon Burgos, Benjamin Charlier, Anne Bertrand, et al. Ad course map charts alzheimer's disease progression. *Scientific Reports*, 11(1):8020, 2021.

[3] Igor Koval, Thomas Dighiero-Brecht, Allan J Tobin, Sarah J Tabrizi, Rachael I Scahill, Sophie Tezenas du Montcel, Stanley Durrleman, and Alexandra Durr. Forecasting individual progression trajectories in huntington disease enables more powered clinical trials. *Scientific Reports*, 12(1):18928, 2022.

[4] Estelle Kuhn and Marc Lavielle. Maximum likelihood estimation in nonlinear mixed effects models. *Computational statistics & data analysis*, 49(4):1020–1038, 2005.

[5] Etienne Maheux, Igor Koval, Juliette Ortholand, Colin Birkenbihl, Damiano Archetti, Vincent Bouteloup, Stéphane Epelbaum, Carole Dufouil, Martin Hofmann-Apitius, and Stanley Durrleman. Forecasting individual progression trajectories in alzheimer's disease. *Nature Communications*, 14(1):761, 2023.

[6] Tim P Morris, Ian R White, and Michael J Crowther. Using simulation studies to evaluate statistical methods. *Statistics in medicine*, 38(11):2074–2102, 2019.

[7] Juliette Ortholand, Stanley Durrleman, and Sophie Tezenas du Montcel. A joint spatiotemporal model for multiple longitudinal markers and competing events. *arXiv preprint arXiv:2501.08960*, 2025.

[8] Dimitris Rizopoulos. *Joint models for longitudinal and time-to-event data: With applications in R*. CRC press, 2012.

[9] Jean-Baptiste Schiratti, Stéphanie Allassonnière, Olivier Colliot, and Stanley Durrleman. A bayesian mixed-effects model to learn trajectories of changes from repeated manifold-valued observations. *Journal of Machine Learning Research*, 18(133):1–33, 2017.